

HCF and LCM

Simple Notes, Methods, Examples & Q\&A

HCF and LCM are two important ideas based on the factors and multiples of numbers. This guide explains both in simple words with examples, the main methods to find them, their relationship, and a set of solved questions.

1. Quick Recap: Factors & Multiples

Factor: A number that divides another number exactly (no remainder). Factors of 12 = 1, 2, 3, 4, 6, 12.

Multiple: The result of multiplying a number by 1, 2, 3, ... Multiples of 4 = 4, 8, 12, 16, 20 ...

2. HCF (Highest Common Factor)

Definition: The largest number that divides two or more numbers exactly. It is also called GCD (Greatest Common Divisor).

Example: HCF of 12 and 18:

- $12 = 2 \times 2 \times 3$
- $18 = 2 \times 3 \times 3$
- Common factors = $2 \times 3 = 6$, so HCF = 6

Notes:

- HCF is always less than or equal to the smallest number.
- HCF of two co-prime numbers (like 8 and 9) is 1.

3. LCM (Least Common Multiple)

Definition: The smallest number that is exactly divisible by two or more numbers (the smallest common multiple).

Example: LCM of 4 and 6:

- Multiples of 4 = 4, 8, 12, 16, 20 ...
- Multiples of 6 = 6, 12, 18, 24 ...
- Smallest common multiple = 12, so LCM = 12

Notes:

- LCM is always greater than or equal to the largest number.

4. Methods to Find HCF & LCM

(a) Prime Factorisation Write each number as a product of primes.

- HCF = product of the COMMON prime factors (lowest powers).
- LCM = product of ALL prime factors (highest powers).
- Example: $12 = 2^2 \times 3$ and $18 = 2 \times 3^2$ $\rightarrow$ HCF = $2 \times 3 = 6$, LCM = $2^2 \times 3^2 = 36$

(b) Division Method (for HCF) Divide the bigger number by the smaller, then divide the divisor by the remainder, and repeat until the remainder is 0. The last divisor is the HCF.

(c) Common Division (for LCM) Divide all numbers together by common prime numbers step by step, then multiply all the divisors.

5. Relationship Between HCF and LCM

For any two numbers, this rule is always true:

$$\text{HCF} \times \text{LCM} = \text{First number} \times \text{Second number}$$

Example for 12 and 18:

$$\text{HCF} \times \text{LCM} = 6 \times 36 = 216 \quad \text{and} \quad 12 \times 18 = 216 \quad (\text{both equal})$$

6. HCF vs LCM (Key Difference)

- HCF deals with FACTORS; LCM deals with MULTIPLES.
- HCF is the GREATEST common factor; LCM is the LEAST common multiple.
- $\text{HCF} \leq$ smallest number; $\text{LCM} \geq$ largest number.

7. Questions & Answers

Q1. Find the HCF of 12 and 18.

- Ans: 6 (common factors 2×3)

Q2. Find the LCM of 4 and 6.

- Ans: 12 (smallest common multiple)

Q3. Find the HCF of 24 and 36.

- Ans: 12 ($24 = 2^3 \times 3$, $36 = 2^2 \times 3^2$, common = $2^2 \times 3 = 12$)

Q4. Find the LCM of 8 and 12.

- Ans: 24 ($8 = 2^3$, $12 = 2^2 \times 3$, $\text{LCM} = 2^3 \times 3 = 24$)

Q5. The HCF of two numbers is 6 and their LCM is 36. If one number is 12, find the other.

- Ans: $(\text{HCF} \times \text{LCM}) / 12 = (6 \times 36) / 12 = 216 / 12 = 18$

Q6. Find the LCM of 5, 10 and 15.

- Ans: 30

Q7. Find the HCF of 16 and 24.

- Ans: 8

Q8. Two bells ring at intervals of 6 and 8 minutes. After how many minutes will they ring together?

- Ans: $\text{LCM}(6, 8) = 24$ minutes.

Q9. Find the greatest number that divides both 20 and 25 exactly.

- Ans: $\text{HCF}(20, 25) = 5$

Q10. Find the LCM of 9 and 12.

- Ans: 36 ($9 = 3^2$, $12 = 2^2 \times 3$, $\text{LCM} = 2^2 \times 3^2 = 36$)

Memory tip: HCF -> Highest common Factor -> small answer (a factor).

LCM -> Least common Multiple -> big answer (a multiple).